

CHE 396 : Senior Design I

MINI- PROJECT 6

12/8/2019

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Project Charter

Title	Goal	In Scope	Out of Scope
MINI- PROJECT-6	To investigate safety and environmental hazard of this process	• External impact • Safety features • Hazard preparation • HAZOP study	• Capital Investment • Location Impact
Deliverables		Team Members	
• Design Plan • Case Description • For R-100 and T-100 - o External impact - o Safety features - o Hazard preparation - o HAZOP study		<i>Gewargis Garmooos</i> <i>Mike Simon</i> <i>Yusuf Sorunke</i> <i>Yohannes Teweldeberhan</i>	

Design Plan

Hazard Analysis and Environmental Assignment					
Task ID	Task	Start Date	Duration	Finish Date	Assignee
Logistics					
1	Organization Task Assignments	12/2/19	1	12/2/19	All
Execution					
3	MSDS Research	12/2//19	2	12/3/19	All
4	Environmental Research	12/3/19	2	12/4/19	Yohannes Yusuf
5	Process Research	12/3/19	2	12/4/19	Gewargis Mike
6	Environmental Analysis	12/4/19	4	12/8/19	Yohannes Yusuf
7	HAZOP Analysis	12/4/19	4	12/8/19	Gewargis Mike
8	Material and Process Hazards	12/4/19	4	12/8/19	Gewargis Mike
Wrap Up					
9	Formatting and Editing	12/8/19	1	12/8/19	All

Executive Summary

In this Mini-Project exercise, the aim is to investigate and ascertain any safety and/or possible environmental hazards inherent or that may accompany vinyl chloride synthesis. Undertaking this task becomes a necessary step so as to study and understand any necessary unavoidable hazards. In the case of fires, release of toxic chemicals and/or explosions, this analysis will help mitigate the resulting damage to both the facility and the environment. To carry this task, moral and ethical obligations as well as law requirements (such as Occupational Safety and Health Act (OSHA), Emergency Planning and Community Right-To-know Act (EPCRA) and Toxic Substance Control Act (TSCA) to mention a few) were considered.

In this case, the Process considered is Vinyl Chloride Production with emphasis on the Direct Chlorination and vinyl Chloride recovery column. All Possible materials Hazards and how certain design feature that can be put in place to mitigate these hazards were considered. Also taken into consideration are the measures put in place in case of any occurrence.

In conclusion, it is important to state that the Possible Hazardous scenarios described subsequently are few when compared to reality and so it is very important to have a very robust safety analysis in place to help, hence saving a company time and money while ensuring safe working conditions for its employees and its immediate surrounding community.

Process And Component Description - Reactor

Direct Chlorination Reactor, R-100

In order to investigate what safety issues one might encounter around the direct ethylene chlorination reactor, first it is important to understand what happens in the reactor. In this reactor, both ethylene and chlorine are introduced into the reactor in gaseous state in separate streams, they are dissolved and diffused, the resulting reaction forms a liquid phase ethylene dichloroethane. With all the possible components of the unit operation known, the Materials Safety Data Sheet for all these chemicals was studied and the findings and the corresponding mitigation measures are discussed subsequently.

Ethylene

At ambient temperature, ethylene has a sweet odor. It has no color, non-irritating and non-corrosive. When not in use, it is cooled and pressurized as a liquid in the industry. It has a normal boiling and freezing point of $-155\text{ }^{\circ}\text{F}$ ($-104\text{ }^{\circ}\text{C}$) and $-272\text{ }^{\circ}\text{F}$ ($-169\text{ }^{\circ}\text{C}$) respectively. Workplace exposure and environmental release of this chemical compound occurs very rarely because of very robust industry guidelines and government requirements. The American Conference of Government Industrial Hygienists (ACGIH) adopts a Threshold Limit Value (TLV) of 200 ppm for ethylene as an 8-hour time-weighted average (TWA) and the typical air levels significantly $<1\text{ ppm}$. [1]

Due to its high vapor pressure, in the case its released, it tends to accumulate in the atmosphere and it is considered slightly toxic to aquatic life in small doses. It is extremely flammable as gas and pressurized. Decomposition with vigorous pressure build-up can occur if stored under conditions above $180\text{ }^{\circ}\text{C}$ and 150 psig. It is highly reactive with oxidizing materials, mineral acids, metals and their chlorides. At high temperatures and/or if a free radical initiator is present, hazardous polymerization can occur.

Chlorine

Chlorine is a greenish-yellow gas with a pungent odor similar to that of bleach with an odor threshold . It is slightly soluble in water forming hypochlorous and hypochloric acid. It is approximately 2.5 times heavier than air. In liquid form, it will evaporate into the air very rapidly and it also combines very readily with all gaseous substances with the exception of rare gases and nitrogen. By itself, Chlorine is not flammable but given that it is highly readily reactive, it can form explosive compounds when it comes in contact with substances such as hydrogen, ammonia and natural gas to name a few. Similar to Ethylene gas, when not in use, chlorine is cooled and pressurized, turning it into liquid with a melting point of -100.98°C and a boiling point of -34.6°C .

Dichloroethane

This compound is liquid at 80°C and colorless . It has a sweet odor that can be detected but can not be classified as a respiratory sensitiser. During a skin contact it may cause skin irritation. In a prolonged exposure it may cause cancer. If swallowed it may cause poisoning effect to central nervous system. The vapor form of this compound may cause a risk of ignition at flash point of 12°C , that is why it is designated as flammable. Since it is heavier than air, it spreads along the floor and forms an explosive mixture with air. But kept in normal ambient and anticipated storage it is stable. Neoprene and Polyethylene are some materials which are not suitable for production. This chemical reacts violently with strong oxidizers.

Mitigation

Measures put in place to prevent or minimize the effect these chemicals might have on people and the immediate environment are very similar in so many ways. For most chemicals, if possible, enclose operations and use local exhaust ventilation at the site of chemical exposure or release.[3] If not, use Respirators should be deployed. Permanently installed analyzers should be in place to continuously monitor the gas chemical release. Hazard And warning information should be posted in visible areas of the workplace and also, ongoing education and training to all personnel.

Material and Process Hazards:

Ethylene and dichloroethane both have flash points below ambient temperatures, creating fire hazards. Chlorine gas has been identified by OSHA as a highly hazardous chemical, creating a toxicity hazard. Both of these flammability and toxicity hazards need to be considered in the event of a loss of containment from reactor R-100 and that regular maintenance and training should focus on preventing loss of containment events.

The fact that the reaction of ethylene and chlorine is an exothermic reaction leads to several different important process hazards associated with the R-100 reactor. If the cooling water feed were to somehow fail, then an energy build up in the reactor would lead to rising temperatures and pressures. If the pressure rises high enough, then the release valve should open, which will lead to a loss of containment if it is not connected to a secondary containment vessel. This loss of containment would create toxicity and fire hazards throughout the plant

If the cooling water failed and the pressure release valve were to also fail, then the rising reactor pressure could lead to an explosion. This would also cause a loss of containment that would create the same toxicity and fire hazards in addition to the explosive blast.

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Based on OSHA regulations, the plant operators need to be made aware of these potential hazards and operating procedures should be established for shutting down the reactor in the event of cooling water failure. The reactor pump that transports the reactor effluent to the furnace is also a potential noise hazard and should be evaluated to determine if the operators will require ear protection in the surrounding area.

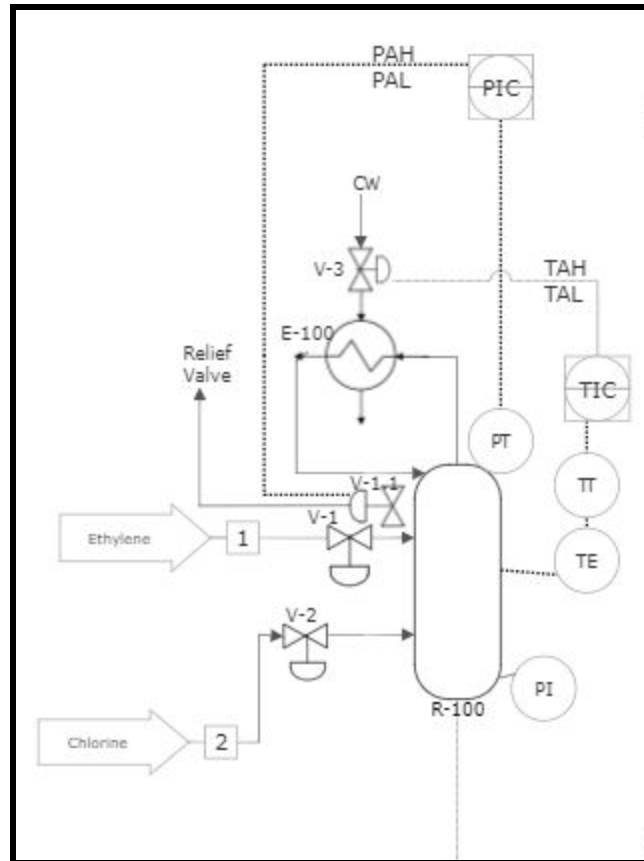


Figure 1: Reactor R-100 After Safety Analysis

Table 1: Reactor R-100 HAZOP Analysis

R-100 <i>Intention - Chlorinates ethylene feed</i>			
Stream S1			
Guide Word	Deviation	Cause	Consequence & Action
No	Flow	Feed valve fail closed	Reactor temperature and pressure fall; TAL and PAL alarms sound
Less	Flow	Ratio control failure	Temperature fall; TAL alarm
As well as	Air Flow	Leak in ethylene storage tank or piping	Possible dangerous unintended reactions; ensure proper maintenance
Reverse	Flow	Pump P-100 fails and reactor level rises	Dichloroethane contaminates ethylene storage; ensure proper maintenance
Stream 2			
Guide Word	Deviation	Cause	Consequence & Action
No	Flow	Feed valve fail closed	Reactor temperature and pressure fall; TAL and PAL alarms sound
Less	Flow	Ratio control failure	Temperature fall; TAL alarm
As well as	Air Flow	Leak in chlorine storage tank or piping	Possible dangerous unintended reactions; ensure proper maintenance
Reverse	Flow	Pump P-100 fails and reactor level rises	Dichloroethane contaminates chlorine storage; ensure proper maintenance
Stream S3			
Guide Word	Deviation	Cause	Consequence & Action
No	Flow	Pump P-100 failure	Reactor pressure increases; PAH alarm sounds; pressure release valve opens
Less	Flow	Pump P-100 failure	Same as “No Flow”
More	Flow	Leak in reactor or piping	Reactor pressure decreases; PAL alarm sounds
As well as	Water Flow	Leak in the heat exchanger	Possible dangerous unintended reactions; ensure proper maintenance
Reverse	Flow	Recycle stream S16 is over-pressurized	Same as “No Flow”

Process And Component Description - Distillation Tower

Vinyl Chloride Recovery Distillation Column, T-100

In order to understand its overall safety of the chemical component in the distillation column, each of the chemicals should be studied based on the MSDS specification. In this distillation column we have 1,2 dichloroethane, Vinyl Chloride and Hydrogen Chloride, the latter one present in both liquid and vapor form but for the rest only in vapor form. This distillation is operating at 12 atm pressure and the rest of the incoming streams are operating at temperature range from -26.4 to 94.6 °C. The hydrogen chloride that is coming at -26.4 °C is the only stream coming in a vapor format.

Vinyl Chloride[7]

This chemical is very flammable having an NFPA rating at 4, in times small tear and wear a leak may cause an explosion at flash point of -78 °C. It is a colourless compound which exists in gas form and that means it is hard to detect using our naked eyes. Even though it can be detected by smell because of its sweet odor, it may cause irregular heartbeats to lung congestion. In case of contact to skin irritation can occur immediately and severe damage to eye in case of contact. During accidental ingestion to this compound a frostbite may occur immediately and if this persists one may be susceptible to cancer.

When looking at the reactivity of the component, it may polymerize unless secured well. Hence it should be protected against light or heat material. It decomposes itself in the presence of halogenated compounds and oxides of carbon. When exposed to metal carbide, metal oxidising material and peroxide it may cause corrosion.

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Hydrogen Chloride [6]

This chemical compound unlike Vinyl chloride it is not flammable and stable under normal conditions. But like Vinyl it is colourless and it can be detected by its pungent irritating smell, and if inhaled heavily it may damage to the mucous membrane in the nose, throat, lungs and bronchial system. It exists in liquid form and gaseous form at 50.5 °C. Since this compound is miscible with water, during undetected leakage it may contaminate clean water bodies and can be harmful to aquatic organisms.

Its corrosive nature causes severe burns during spillage or poor handling. If contacted with eye it lead to severe sight impairment and if ingested it may produce burning effects to the lips, oral cavity upper airway and digestive tract. In General, during prolonged contact it causes a serious tissue damage.

Once again because of its corrosive nature it will corrode when it comes in contact with bases and metals . React vigorously when contacted with amines, reducing agents and organic material.

1,2 Dichloroethane [4]

This compound is liquid at 80 °C and colorless . It has a sweet odor that can be detected but can not be classified as a respiratory sensitiser. During a skin contact it may cause skin irritation. In a prolonged exposure it may cause cancer. If swallowed it may cause poisoning effect to central nervous system. The vapor form of this compound may cause a risk of ignition at flash point of 12 °C, that is why it designated as flammable. Since it is heavier than air, it spread along floor and form explosive mixture

with air. But kept in normal ambient and anticipated storage it is stable. Neoprene and Polyethylene are some material which is not suitable for production. This chemical reacts violently with strong oxidizers.

Mitigation Steps

Before Spillage

Shutdown: Easy access for local and overall shutdown, incase of emergency.

Distillation material : Leakage of any substance is not an option at any scale. Regular check up inside distillation column for corrosive is necessary. All material that can withstand corrosive nature of chemicals should be selected.

Reboiler: Because of the presence of a compound that has very low flash point temperature, the boiler should have a temperature control device. The pressure that is produced during this heat should be controlled. Alarms should be placed near the boiler.

Condenser: To guarantee continuous cooling of the chemical compound a temperature gauge should be installed. And the pressure valve that control cooling water should be checked and maintained regularly.

Pipe check: To prevent any blockage of pipes due to unwanted residue or corrosive build up, all pipes going through or around the distillation should be flushed regularly.

After Spillage (Containment):

Silica Sand: In time of accidental spillage on the ground a sand should be spread on the liquid material, this would help to absorb the liquid before it reach any aquatic bodies.

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Ventilation: To avoid any unnecessary pressure build up or temperature rise around the distillation column Regular ventilation of the vicinity is absolutely necessary.

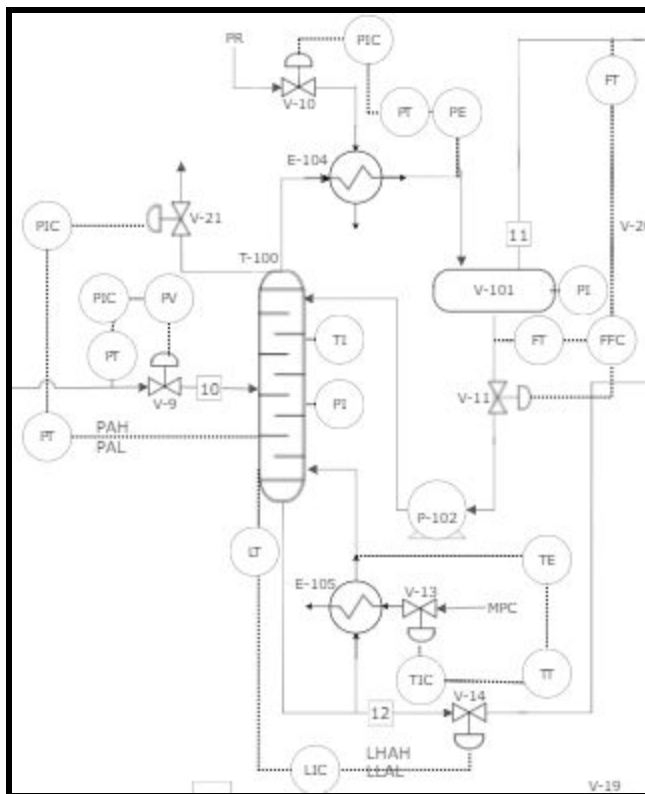


Figure 2: Distillation T-100 Column After Safety Analysis

Materials and Process Hazards

Distillation towers can be considered one of the most exhausting and difficult pieces of equipment to operate because there are so many factors to take into consideration. Temperature, pressure, number of trays are just a few but this process is specifically hazardous, dealing with harsh chemicals such as hydrochloric acid and vinyl chloride. Hydrochloric acid is a very hazardous chemical. Its properties when interacting with materials such as metals, can be corrosive, thus, causing detrimental

effects to the nearby environment. At high temperatures, it will decompose into hydrogen and chlorine. OSHA has identified hydrochloric acid as a highly hazardous chemical.

Chlorine gas in itself can be liquified at $-35\text{ }^{\circ}\text{C}$ in which in this specific process, a temperature of $-26.4\text{ }^{\circ}\text{C}$ will be reached concluding, there can be a possibility of chlorine condensating. Chlorine gas is also highly toxic and has also been identified by OSHA as a highly hazardous chemical.

Vinyl chloride also has its own harsh properties. It is known to cause frostbite by evaporative cooling, and leaks can be either a liquid or gas. With HCl being involved, chances of a fire can be very high as its flash point is low. Under OSHA regulations, VC (vinyl chloride), suspects it to be a possible carcinogen.

In order to avoid these catastrophic events from happening, specific utilities must be working as its peak potential. Valve V-9 must continually supply the same pressure into the distillation tower at 12 atm. Any deviation in the pressure of the column could disrupt the equilibria conditions and cause contamination in the column products that could lead to unintended hazards. Looking back at OSHA regulations, both HCl and VC is very toxic to the human body, prolonged exposure causes respiratory problems and likely death. Pressure relief valves on the distillation column is important to prevent over pressurization hazards, but to prevent HCl loss of containment, a secondary containment vessel from the pressure relief should be considered.

Temperatures inside and around the T-100 distillation column cannot exceed its proper designated source, meaning, if a stream temperature is exceeded, complications will most definitely occur. The highest temperature that the T-100 distillation column will be operating at just below $100\text{ }^{\circ}\text{C}$. A temperature control must also be implemented because the highest temperature near T-100 distillation column is just under $100\text{ }^{\circ}\text{C}$ and the flash point of HCl is just above $100\text{ }^{\circ}\text{C}$, sitting at around $108\text{ }^{\circ}\text{C}$. The lowest temperature operating near T-100 is around $-30\text{ }^{\circ}\text{C}$. Flash point of Vinyl Chloride is at $-78\text{ }^{\circ}\text{C}$.

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Although there is a gap between the two temperatures, further safety additional measures must be implemented. Temperature controls near and around the T-100 distillation column should be used in order for vinyl chloride staying at its intended liquid phase.

The condenser pump that refluxes the condensate to the top of the column is also a potential noise hazard and should be evaluated to determine if the operators will require ear protection in the surrounding area.

Table 2: Distillation Tower T-100 HAZOP Analysis

T-100 <i>Intention - Separates HCl from VC and Dichloroethane</i>			
Feed			
Guide Word	Deviation	Cause	Consequence & Action
No	Flow	Valve V-9 fail	Level drops, LAL alarm sounds
Less	Flow	Valve V-9 fail	Same as “No Flow”
More	Pressure	Valve V-9 fail	Pressure alarm high; pressure release valve opens
Less	Pressure	Valve V-9 fail	Pressure alarm low
As well as	Composition	Refrigerant leaking in E-103	Temperature drops; possible unintended reactions; ensure proper maintenance
Vapor Out			
Guide Word	Deviation	Cause	Consequence & Action
No	Flow	Reboiler fails	V-101 level drops; consider adding a level level alarm or temperature alarm
Less	Flow	Reboiler fails	Same as “No Flow”
As well as	Composition	Reboiler is vaporizing vinyl chloride	V-101 level rises
Condenser In			
Guide Word	Deviation	Cause	Consequence & Action
No	Flow	Pump P-102 or condenser E-104 fails	V-101 level floods; ensure proper maintenance
Less	Flow	Pump P-102 or condenser E-104 fails	Same as “No Flow”
As well as	Composition	Refrigerant leaking in E-104	Temperature drops; possible unintended reactions; ensure proper maintenance

Table 2 (Continued)

Liquid Out			
Guide Word	Deviation	Cause	Consequence & Action
No	Flow	Valve fail	Column level rises; LAH alarm sounds
Less	Flow	Valve fail	Same as “No Flow”
More	Flow	Tower, reboiler, or piping leaking	Column level drops; LAL alarm sounds
Boilup In			
Guide Word	Deviation	Cause	Consequence & Action
No	Flow	Reboiler E-105 fails	Temperature drops; consider adding a temperature alarm
Less	Flow	Reboiler E-105 fails	Same as “No Flow”
As well as	Composition	Steam leaking from E-105	Possible unintended reactions; ensure regular maintenance

References

- [1] “Ethylene” Product Safety Assessment Document www.dow.com/productsafety/finder/
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